RCS04 | Instructor Guide

CTS Cohort 6 | CTS Day 12 | 2026-07-23 | 10:00 AM-3:00 PM | 6 students

Plumbing Basics: Supply, Drainage, and Dry Verification

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Big idea: Water follows pressure and gravity; reliable plumbing depends on correct parts, alignment, slope, seals, access, and verification.

Architecture of the day

Time	Instructor move
10:00-10:15	Arrival, PPE, retrieval warm-up
10:15-10:30	Morning quiz
10:30-10:50	Quiz review and system-path check
10:50-11:20	Supply and DWV system mapping
11:20-12:00	P-trap parts, washer, and alignment demonstration
12:00-12:30	Dry-verification guided practice and lab release
12:30-1:00	Lunch and controlled reset
1:00-2:25	Rolling-sink module rebuild and capstone pathway lab
2:25-2:45	Fault-find, dry recheck, evidence, cleanup
2:45-3:00	Exit ticket and Day 5 preview

Learning targets

- Differentiate supply and drain-waste-vent functions
- Explain fixture, trap, and shutoff roles
- Plan a training plumbing path through framing
- Assemble and troubleshoot a dry rolling-sink/P-trap module

Materials and setup

- Rolling-sink and P-trap training modules with complete parts inventories
- Tailpieces, trap arms, slip nuts, correctly sized washers, and replacement parts
- Supply/DWV system maps and capstone coordination drawings
- Dry inspection trays, towels, and unexpected-moisture controls
- Separate plumbing work zone with no electrical cords or test sources
- Evidence camera or approved device

Instructor talking points

- Use RCS01 rolling-sink experience as retrieval, then deepen system mapping.
- Trace supply -> shutoff -> fixture and fixture -> tailpiece -> P-trap -> drain outlet.
- Explain trap seal purpose without claiming a dry classroom mock-up proves code compliance.
- Demonstrate washer orientation, hand-tight alignment, accessible joints, dry inspection, and one-change-at-a-time troubleshooting.
- Coordinate plumbing paths with framing and electrical zones using the approved training drawing.

Whiteboard plan

1. Supply path
2. Drain path
3. P-trap parts
4. Alignment before tightening
5. Inspect -> find one issue -> correct -> recheck

Lab facilitation

Teams rebuild and troubleshoot rolling sink/P-trap modules using dry verification only, then each student maps a coordinated plumbing pathway on the capstone wall without making unapproved structural penetrations.

Lab sequence

6. Inventory module
7. Lay out system path
8. Inspect parts
9. Dry-fit
10. Orient washers
11. Hand assemble
12. Alignment inspection
13. Instructor inserts or identifies one fault
14. Diagnose one issue at a time
15. Correct and dry-reverify
16. Map capstone path
17. Document and reset

Safety boundaries

- No water or pressure testing unless the instructor explicitly authorizes and controls it
- No electrical tools or cords in the plumbing station
- Wipe any unexpected moisture immediately

- No structural penetration without instructor approval
- Stop for cracked, missing, or cross-threaded parts

Evidence targets

- System map
- P-trap parts identification
- Dry assembly inspection
- Troubleshooting record
- Coordinated capstone pathway sketch

Quiz answer key

1. What system delivers water to a fixture? The supply system.

2. What does DWV stand for? Drain, waste, and vent.

3. What is the P-trap's basic purpose? To hold a water seal that helps block sewer gases.

4. What should happen before tightening slip joints? Parts and washers should be aligned and correctly oriented.

5. Why troubleshoot one change at a time? So the crew can identify which correction changed the result.

6. What does a shutoff valve allow? Isolation of the fixture supply.

7. May students drill framing for plumbing without approval? No.

8. What type of verification is planned for today's lab? Dry visual and mechanical verification.

9. Why keep plumbing and electrical stations separated? To prevent moisture from creating electrical hazards.

10. What does a professional do when a part is cracked or cross-threaded? Stop, identify it, and replace or correct it rather than force it.

Exit-ticket look-fors

- Draw the drain path
- Name one assembly fault
- Describe the dry-verification boundary
- Explain one framing/plumbing coordination choice

Google Classroom post

RCS Day 4 - Plumbing Basics: Supply, Drainage, and Dry Verification

Today we focused on water follows pressure and gravity; reliable plumbing depends on correct parts, alignment, slope, seals, access, and verification. Students completed guided instruction,

hands-on capstone work, verification, evidence capture, and cleanup. Submit your worksheet, lab evidence, and exit ticket before leaving.

Source and safety references

- **Canonical RCS skeleton:** /CANONICAL/00_INBOX/Downloads/rcs_curriculum_plan.md
- **Cohort 6 RCS01 source:** /CANONICAL/02_PROJECTS/CTS_Cohort_06/03_DAILY/D09_RCS01_2026-07-20/Source/lesson.md
- **OSHA PPE:** https://www.osha.gov/personal-protective-equipment/construction
- **OSHA eye and face protection:** https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.102
- **OSHA circuit lockout/tagging:** https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.417
- **Oregon electrical codes:** https://www.oregon.gov/bcd/codes-stand/pages/electrical.aspx
- **Oregon permits:** https://www.oregon.gov/bcd/lbdd/pages/oregon-permits.aspx

RCS04 | Morning Quiz - Answer Key

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Plumbing Basics: Supply, Drainage, and Dry Verification

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The supply system.

2. What does DWV stand for?

Drain, waste, and vent.

3. What is the P-trap's basic purpose?

To hold a water seal that helps block sewer gases.

4. What should happen before tightening slip joints?

Parts and washers should be aligned and correctly oriented.

5. Why troubleshoot one change at a time?

So the crew can identify which correction changed the result.

6. What does a shutoff valve allow?

Isolation of the fixture supply.

7. May students drill framing for plumbing without approval?

No.

8. What type of verification is planned for today's lab?

Dry visual and mechanical verification.

9. Why keep plumbing and electrical stations separated?

To prevent moisture from creating electrical hazards.

10. What does a professional do when a part is cracked or cross-threaded?

Stop, identify it, and replace or correct it rather than force it.

Scoring guidance

One point per item. Accept technically equivalent wording. Use errors to reteach before lab release. Safety-boundary errors must be corrected explicitly before the student begins the related task.

RCS04 | Student Packet

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Learning targets

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- Explain fixture, trap, and shutoff roles
- Plan a training plumbing path through framing
- Assemble and troubleshoot a dry rolling-sink/P-trap module

Today's run

10:00-10:15 | Arrival, PPE, retrieval warm-up

10:15-10:30 | Morning quiz

10:30-10:50 | Quiz review and system-path check

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12:30-1:00 | Lunch and controlled reset

1:00-2:25 | Rolling-sink module rebuild and capstone pathway lab

2:25-2:45 | Fault-find, dry recheck, evidence, cleanup

2:45-3:00 | Exit ticket and Day 5 preview

Key words

supply | shutoff | fixture | tailpiece | P-trap | trap arm | DWV | slope | washer | slip nut | cleanout

Guided worksheet

1. Draw the supply and drain paths.

_____

_____

2. Label the P-trap, tailpiece, trap arm, and slip-joint washer.

_____

_____

3. Explain why cross-threading causes failure.

_____

_____

4. Mark a safe training path that avoids the electrical boxes.

_____

_____

5. Write the dry-verification sequence.

_____

_____

Afternoon lab mission

Teams rebuild and troubleshoot rolling sink/P-trap modules using dry verification only, then each student maps a coordinated plumbing pathway on the capstone wall without making unapproved structural penetrations.

Safety boundary

- No water or pressure testing unless the instructor explicitly authorizes and controls it
- No electrical tools or cords in the plumbing station
- Wipe any unexpected moisture immediately
- No structural penetration without instructor approval
- Stop for cracked, missing, or cross-threaded parts

Build / test sequence

[ ] Inventory module

[ ] Lay out system path

[ ] Inspect parts

[ ] Dry-fit

[ ] Orient washers

[ ] Hand assemble

[ ] Alignment inspection

[ ] Instructor inserts or identifies one fault

[ ] Diagnose one issue at a time

[ ] Correct and dry-reverify

[ ] Map capstone path

[ ] Document and reset

Evidence record

[ ] System map: _____

[ ] P-trap parts identification: _____

[ ] Dry assembly inspection: _____

[ ] Troubleshooting record: _____

[ ] Coordinated capstone pathway sketch: _____

Reflection and exit ticket

Draw the drain path

_____

_____

Name one assembly fault

_____

_____

Describe the dry-verification boundary

_____

_____

Explain one framing/plumbing coordination choice

_____

_____

Instructor verification

Status: [ ] Approved [ ] Correct and recheck [ ] Needs follow-up

Instructor initials: _____ Date/time: _____

RCS04 | Morning Quiz

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Plumbing Basics: Supply, Drainage, and Dry Verification

Name: _____ Date: _____

Answer from your own understanding. This is a teaching check, not a trap.

1. What system delivers water to a fixture?

_____

_____

2. What does DWV stand for?

_____

_____

3. What is the P-trap's basic purpose?

_____

_____

4. What should happen before tightening slip joints?

_____

_____

5. Why troubleshoot one change at a time?

_____

_____

6. What does a shutoff valve allow?

_____

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7. May students drill framing for plumbing without approval?

_____

_____

8. What type of verification is planned for today's lab?

_____

_____

9. Why keep plumbing and electrical stations separated?

_____

_____

10. What does a professional do when a part is cracked or cross-threaded?

_____

_____